ad_antiab_kg

A triangulated multi-modal knowledge-graph, heterogeneous graph transformer (HGT) and Mendelian randomization framework for ARIA risk stratification and drug-repurposing screening in anti-amyloid-β immunotherapy for Alzheimer's disease

Companion code, knowledge graph, fusion tables and supplementary materials for the manuscript submitted to *Pharmacological Research*.

1. Overview

This repository contains all analysis code, the constructed knowledge graph (KG node/edge tables), the intermediate data-fusion tables and the Supplementary Materials supporting the study. The pipeline integrates five open-access resources:

Data source	Role in the study	Chapter
FDA FAERS (openFDA API)	Disproportionality signal detection (ROR / PRR / BCPNN / MGPS) for ARIA-related events	Ch. 1–2
ClinicalTrials.gov	Anti-Aβ trial information and ARIA reporting	Ch. 1–2
DrugBank / ChEMBL / GO / Reactome / KEGG	KG node and edge construction	Ch. 3

OpenGWAS (IEU OpenGWAS)	Genetic instruments for 24 genes and 9 outcome GWAS	Ch. 5
eQTL Catalogue (EMBL-EBI)	Brain and blood eQTL for colocalization	Ch. 5

The framework follows a **triangulation logic**: (i) pharmacovigilance signal detection, (ii) mechanistic KG + graph neural network (GNN/HGT) risk prediction, and (iii) causal inference by two-sample Mendelian randomization (MR) and Bayesian colocalization, finally fused into a single ranked list of repurposing candidates and combination partners.

2. Repository structure

```
ad_antiab_kg/
├── README.md
├── LICENSE
├── CITATION.cff
├── environment.yml              # Python 3.10 (torch 2.3.1, torch-geometric 2.5.3, ...)
├── requirements_R.txt           # R package list (TwoSampleMR, coloc, ...)
├── src/
│   ├── py/
│   │   ├── 01_download_faers.py       # Ch.2  FAERS download + cleaning
```

```
|   |   ├── 03_kg_build.py            # Ch.3  multi-modal KG
construction
|   |   ├── 04_gnn_risk.py            # Ch.4  GNN risk prediction
|   |   ├── 08_gnn_linkpred_hgt.py    # Ch.6  HGT link prediction
(train / predict)
|   |   └── 08b_fusion_v3.py          # Ch.6  multi-evidence fusion &
ranking
|   └── r/
|       ├── 02_signal_detection.R     # Ch.2  ROR / PRR / BCPNN /
MGPS
|       ├── 05_mr_validation.R        # Ch.5  two-sample MR
|       ├── 06_coloc.R                # Ch.5  colocalization
(blood/generic)
|       └── 07_coloc_brain.R          # Ch.5  colocalization (brain
eQTL)
├── data/
|   ├── kg/          nodes.parquet, edges.parquet      # constructed
KG (core)
|   ├── kg_ext/      nodes.parquet, edges.parquet      # extended
KG (HGT input)
|   └── (faers/, raw/, interim/  NOT included - large raw data,
regenerate with src scripts)
```

├── results/

│ ├── tables/ signal tables, MR summary, fusion ranking, paper tables

│ └── figures/ generated figures

└── supplementary/ Table_S1-S4, FigS1, Excel compilation

NOTE. All scripts were written for a portable layout. Original absolute paths

(e.g. /home/<user>/ad_antiab_kg) must be adapted to your own environment;

most Python scripts accept --proj . to define the project root.

3. Environment

Python (3.10)

conda env create -f environment.yml

conda activate neuroai

Key packages: torch==2.3.1, torch-geometric==2.5.3, pandas, numpy, scipy, networkx, pyarrow, lifelines, matplotlib.

R (4.4)

install.packages(c("remotes","BiocManager"))

install.packages("TwoSampleMR")

install.packages("coloc") # v5.2.3

Recommend installing R packages from conda-forge binaries if compiling from

source fails, e.g. conda install -c conda-forge r-coloc.

4. Reproducing the analysis

```
# Ch. 2 - pharmacovigilance signals
python src/py/01_download_faers.py
Rscript src/r/02_signal_detection.R
# Ch. 3 - knowledge graph
python src/py/03_kg_build.py --proj .
# Ch. 4 - GNN drug-risk prediction
python src/py/04_gnn_risk.py --proj .
# Ch. 5 - causal inference
Rscript src/r/05_mr_validation.R
Rscript src/r/06_coloc.R
Rscript src/r/07_coloc_brain.R
# Ch. 6 - HGT link prediction and evidence fusion
python src/py/08_gnn_linkpred_hgt.py predict --proj . --device cuda
python src/py/08b_fusion_v3.py --proj .
```

Outputs are written to results/tables/ (CSV) and results/figures/ (PNG/SVG).

Long-running steps are idempotent and resumable.

5. Supplementary Materials

File	Content
Table_S1_421_drugs_full_risk.csv	Full risk table for all 421 screened drugs (25 columns: GNN scores, FAERS ROR, MR evidence, liability grade, final tier, notes)
Table_S2_HGT_training_curves.csv	Per-epoch HGT training/validation metrics
Table_S3_full_MR_results.csv	Complete MR results (24 genes x 9 outcomes; nSNP, beta, SE, p, OR, 95% CI, Egger, MR-PRESSO, Q, I^2, F, Steiger)
Table_S4_colocalization.csv	Full colocalization results (6 brain regions + blood; PP.H0–H4)
FigS1_HGT_training_curves.png	HGT 50-epoch training curves
Paper_Supplementary_Tables.xlsx	Excel compilation of the above + main Tables 1–3

6. Data availability

All primary data are publicly available: FDA FAERS via the openFDA API, ClinicalTrials.gov API, OpenGWAS (IEU), eQTL Catalogue (EMBL-EBI), DrugBank, ChEMBL, GO, Reactome and KEGG. Code, constructed KG and derived tables generated in this study are provided in this repository.

7. Citation

If you use this code or data, please cite:

[Author list]. [Title]. *Pharmacological Research* (2026). DOI: [to be assigned]

A machine-readable citation is available in CITATION.cff.

8. License

Code is released under the **MIT License** (see LICENSE).

Derived data tables are released under **CC BY 4.0**.

9. Contact

Corresponding author: [name], [institution], [email]